

Indicative Practical List

1. Write a program to print "Hello World" using echo
2. Write a program to print "Hello PHP" using variable
3. Write a program to print a string using echo and variable.
4. Write a PHP program to add two numbers
5. Write a program to print two variables in single echo
6. Write a program to check student grade based on marks
Conditions:
If marks are 60% or more, grade will be First Division.
If marks between 45% to 59%, grade will be Second Division.
If marks between 33% to 44%, grade will be Third Division.
If marks are less than 33%, student will be Fail.
7. Write a program to calculate Electricity bill in PHP
Conditions:
For first 50 units – Rs. 3.50/unit
For next 100 units – Rs. 4.00/unit
For next 100 units – Rs. 5.20/unit
For units above 250 – Rs. 6.50/unit
8. Write a PHP program to check if a person is eligible to vote
9. Write a simple calculator program in PHP using switch case
10. Write a program to show day of the week using switch
11. Write a factorial program using for loop in php
12. Write a program to count 5 to 15 using PHP loop
13. Write a program to create Chess board in PHP using for loop
14. Write a Program to create given pattern with * using for loop
15. Remove specific element by value from an array in PHP?
16. Write a PHP program to calculate area of rectangle
17. Write a PHP program to check whether a number is positive, negative or zero
18. Write a PHP program to reverse the string
19. Write a PHP program to find the length of the string
20. Write a PHP program to count the words in the string
21. Write a PHP program to convert a string into uppercase
22. Create a HTML form that accept the user name and display the name using php
23. Write a PHP script to get the client IP address



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24. Write a simple PHP browser detection script
25. Write a PHP script, which changes the color of the first character of a word
26. Write a PHP script to redirect a user to a different page
27. Write a simple PHP program to check that email id is valid or not
28. Write a PHP script to display string, values within a table
29. Write a PHP function to test whether a number is greater than 30, 20 or 10 using ternary operator
30. Write a PHP program to swap two variables
31. Write a PHP program to check if a number is an Armstrong number or not
32. Write a PHP program to convert word to digit
33. Write a PHP program to remove duplicates from a sorted list
34. Write a PHP program to compute the sum of the prime numbers less than 200
35. Write a PHP program to print out the multiplication table upto 5*5
36. Write a PHP program to compute the sum of the digits of a number
37. Write a PHP program that multiplies corresponding elements of two given lists
38. Write a PHP script which will display the Cars in the following way using array
39. Write a PHP script to get the Second element of the above array
40. Write a PHP script that inserts a new item in an array in any position
41. Write a PHP script to calculate and display average temperature, five lowest and highest temperatures
42. Write a PHP script which displays all the numbers between 50 and 150 that are divisible by 4
43. Write a PHP script to get the shortest/longest string length from an array
44. Write a PHP script to generate unique random numbers within a range
45. Write a PHP function that returns the lowest integer that is not 0
46. Write a PHP script to sort an array in reverse order (highest to lowest)
47. Write a PHP function to set union of two arrays
48. Write a PHP script that displays 1 2 3 4 5 6 7 8 9 10 numbers on one line
49. Write a PHP script using a for loop to add all the integers between 0 and 20 and display the sum
50. Write a PHP script to calculate and print the factorial of a number using a for loop
51. Write a program which will count the "t" characters in the text "tech study"
52. Write a PHP script using nested for loop that creates a chess board



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53. Write a PHP script to print all even numbers between 1 to 10
54. Write a PHP script to print all odd numbers between 1 to 10
55. Write a PHP Script to sum of even numbers from 1 to 10
56. Write a PHP Script to sum of odd numbers from 1 to 10
57. Write a PHP program to print table of number
58. Write a PHP script to convert lowercase string to uppercase
59. Write a PHP script to convert uppercase string to lowercase
60. Write a PHP script to convert first character uppercase
61. Write a PHP script to convert first character of all the words uppercase
62. Write a PHP script split the following string
63. Write a PHP script to extract the user name from the email ID
64. Write a PHP script to get the last three characters of a string
65. Write a PHP script to replace the first 'the' of the following string with 'best'
66. Write a PHP script to generate simple random password from a given string
67. Write a PHP script to print the next character of a specific character
68. Write a PHP script to remove a part of a string from the beginning
69. Write a PHP script to insert a string at the specified position in a given string
70. Write a PHP script to remove all leading zeroes from a string
71. Write a PHP script to remove comma(s) from the numeric string
72. Write a PHP script to get the first word of a sentence
73. Write a PHP script to print letters from 'a' to 'z'
74. Write a PHP script to select first 6 words from the string
75. Create user registration form with First Name, Last Name, gender, date of birth, contact, address, email, password, confirm password and stream. Put JavaScript validations:
 - All fields are mandatory
 - First name, last name must contain alphabets only.
 - Gender must be selected by 'radio' or 'select'
 - Birth date must be valid and before current date.
 - Contact must contain 10 digits.
 - Email must be valid.
 - Password and confirm password must be same.If any of the validation fails, display appropriate message on the document.
76. Create a login page with two fields email and password. Put JavaScript validations:



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- Both the fields are mandatory.
- Email must be a valid email id
- Password must contain 6-20 characters.

If any of the validation fails, display appropriate message on the document.

77. Write PHP script that accepts data submitted by registration form (program-1). Check validations on server side and display valid data.

- All fields are mandatory
- Email must be valid
- Birth date must be valid and before current date.
- Contact must be of 10 digits
- Password and confirm password must be same.

If any of the validation fails, display appropriate message on “registration.php” page.

78. Create password reset page that allows user to reset password by clicking email link which is sent from program-8.

79. Create MySQL table named ‘files’ (id, userId, filePath). Create PHP script that can upload file and stores file information in ‘files’ table.

- Only .docx and .pdf files are allowed with 5MB maximum size.
- To upload file a user must be logged in.
- The actual file should be stored in user’s directory on server.
- This directory must be generated dynamically in uploads directory.

80. Modify the program to show list of files on the user’s dashboard. Also provide remove button to delete a file.

81. A bookshop has a web page displaying books along with prices available. A User needs select books with required quantities, the selected books should be shown in a shopping cart using cookies.

82. Modify program-12 to allow a user to alter his/her choices from cart and can see the final amount payable.

83. Create a PHP admin page to manage Student data in database (CRUD). This admin panel must be coded in object-oriented manner.

84. Write a PHP code which generates a random password of 8 characters. The password must have at least one number, one special character and one Capital letter.



Flutter Practical Assignment-1

1. Create a Fultter application for demonstrating the basics of Flutter project.
2. Write a program to display “Welcome to Flutter” in the center of the screen. Set up an Android Virtual Device for your project and run your program on this virtual device.
3. Write a program to display “Flutter works on Web!” in the center of the screen. Set up Google Chrome browser as the target device and run your program in the browser.
4. Demonstrating how to create Emulator in Flutter project.
5. Write a program to display at least four built-in icons on the screen.
6. Write a program to display at least four images-each of different type such as JPEG, PNG, GIF etc. Also display an image available on Internet using its URL.
7. Create a Fultter application for demonstrating the images in flutter project.
8. Create a Fultter application for demonstrating the app icons of Flutter project.
9. Replace default app icons for a Flutter project manually.
10. Write a program to print your name in Dart.
11. Write a program to print **Hello I am “John Doe”** and **Hello I’am “John Doe”** with single and double quotes.
12. Write a program in Dart that finds simple interest.

$$\text{Formula} = (p * t * r) / 100$$

13. Write a program to print a square of a number using user input.
14. Write a program to print full name of a from first name and last name using user input.
15. Write a program to find quotient and remainder of two integers.
16. Write a program to swap two numbers.
17. Write a program in Dart to remove all whitespaces from String.
18. Write a dart program to convert String to int.
19. Suppose, you often go to restaurant with friends and you have to split amount of bill. Write a program to calculate split amount of bill. $\text{Formula} = (\text{total bill amount}) /$
 number of people



Flutter Practical Assignment-2

1. Write a dart program to check if the number is odd or even.
2. Write a dart program to check whether a character is a vowel or consonant.
3. Write a dart program to check whether a number is positive, negative, or zero.
4. Write a dart program to print your name 100 times.
5. Write a dart program to calculate the sum of natural numbers.
6. Write a dart program to generate multiplication tables of 5.
7. Write a dart program to generate multiplication tables of 1-9.
8. Write a dart program to create a simple calculator that performs addition, subtraction, multiplication, and division.
9. Write a dart program to print 1 to 100 but not 41.
10. Write a program which takes two numbers as input. Provide four buttons for addition, subtraction, multiplication and division. Display output of the arithmetic operation in another Text.
11. Display a group of five checkboxes and display all the currently selected items in a Text as and whenever any checkbox selected or deselected.
12. Re-write program. 1 by replacing buttons with four radiobuttons for addition, subtraction, multiplication and division.
13. Write a program to create two tabs with different background colors. First tab should display three images with the help of a Row widget and the second tab should display three images with the help of Column widget
14. Write a program to create an app with four screens. Each screen must have three buttons to re-direct to one of the three other screens. Use named routes



Flutter Practical Assignment-3

1. Write a program in Dart to print your own name using function.
2. Write a program in Dart to print even numbers between intervals using function
3. Write a program in Dart that generates random password.
4. Write a program in Dart that find the area of a circle using function.
5. Write a program in a dart that implements the Pythagorean theorem using function.
6. Write a program in Dart to reverse a String using function.
7. Write a program in Dart to calculate power of a certain number. For e.g $5^3=125$
8. Write a program to display three horizontal scrollbars: one for each of Principle, Rate of Interest and Term (i.e. no. of years). Calculate and display simple interest depending on the current value of principle, rate of interest and term.
9. Create a scrollable dynamic list of 25 checkboxes with the help of ListView and builder() method. Create a dynamic grid of three columns. Display seven icons in this grid.
10. Write a program to create a drawer with three options: Login, Greet and Close. Create three dart files: Login.dart, Greet.dart Tapping on Login should display screen for login from Login.dart Tapping on Greet should display screen with "Have a good day" message from Greet.dart. Tapping on Close should close the drawer.

Flutter Practical Assignment -4

1. Create an application to demonstrate stateless and stateful Widget in Flutter.
2. Write a dart code to simulate this UI for Calculator in Flutter.



3. Create an application to demonstrate Row, Column Widget in Flutter.
4. Create an application to demonstrate ElevatedButton Widget in Flutter.
5. Create an application to demonstrate Container, Expanded and Card Widget in Flutter.
6. Create an application to demonstrate Container, Expanded and Card Widget in Flutter.
7. Create an application to implement Material Icons, Theme in Flutter.
8. Create an application to demonstrate Refractor Widget (Stateless and Stateful) in Flutter.



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Flutter Practical Assignment -5

1. Create a list of names and print all names using list.
2. Create a set of fruits and print all fruits using loop.
3. Create a program that reads list of expenses amount using user input and print total.
4. Create an empty list of type string called days. Use the add method to add names of 7 days and print all days.
5. Add your 7 friend names to the list. Use where to find a name that starts with alphabet a.
6. Create a map with name, address, age, country keys and store values to it. Update country name to other country and print all keys and values.
7. Create a map with name, phone keys and store some values to it. Use where to find all keys that have length 4.
8. Create a simple to-do application that allows user to add, remove, and view their task.



Flutter Practical Assignment -6

1. Write a program to read data from a JSON file and display the same using a Flutter app.
2. Write a program to write data to a JSON file using a Flutter app.
3. Write a program to display current location of the device using a Flutter app.
4. Write a program with two screens: The first screen will have an empty text and a button to move to second screen. The second screen will have a textbox and a button to move to the first screen. The value entered in the textbox should be displayed in the text on the first screen.
5. Write a program to perform create, retrieve, update and delete data from an online REST API.

Flutter Practical Assignment -7

1. Write a program to show animation using Hero widget in a Flutter app.
2. Write a program to create and display BottomSheet in a Flutter app.
3. Write a program to display real-time clock using StreamBuild in a Flutter app.
4. Write a program to perform create, retrieve, update and delete operations on data available in Firebase.



Indicative Practical List

Assignment – 1

Sr. No	Questions										
Q1	Create an ASP .NET Application using MVC Architecture following requirements.										
	Application must showcase the techniques of Fetching of data from the database using Connected or Disconnected Approach.										
	Application must showcase the techniques of inserting the data into the database using Connected or Disconnected Approach.										
	Application must have well defined views to provide interaction to its functionalities.										
	Application must also be having functionalities to allow user to delete and update records, through a view.										
	Note: For reference you may use below provided structure to create table in database or it is up to you to select one of your choice.										
<table><tr><th colspan="5">TBLFoods</th></tr><tr><th>ID</th><th>Name</th><th>Cuisine</th><th>Price</th><th>Quantity</th></tr></table>		TBLFoods					ID	Name	Cuisine	Price	Quantity
TBLFoods											
ID	Name	Cuisine	Price	Quantity							

Assignment – 2

Create an ASP .NET WEB API Application to fulfill following requirements.

Questions

1. Create a Web API to fetch all the data from the database and return it. (Use Attribute based routing; to allow consumption of the API, for the consumer)
2. Create a Web API to fetch all the data of specific cuisine, as per the request of consumer.
3. Create a Web API to get data from the consumer and insert it in the Database.
4. Add validation in model with Data Annotations and make Name and Quantity a Required Field, create a custom exception to generate a proper error message and avoid data insertion, if the same is not followed by the consumer while consuming the “Data Insert API” (3).
5. While inserting the data in “Data Insert API” (3), check if Quantity is not less than 0, for such case; create a custom exception to return proper message and avoid data insertion.

Note: Use the provided structure to create tables in database.

TBLFoods				
ID	Name	Cuisine	Price	Quantity

Note: Use Entity Framework for Database Operations.